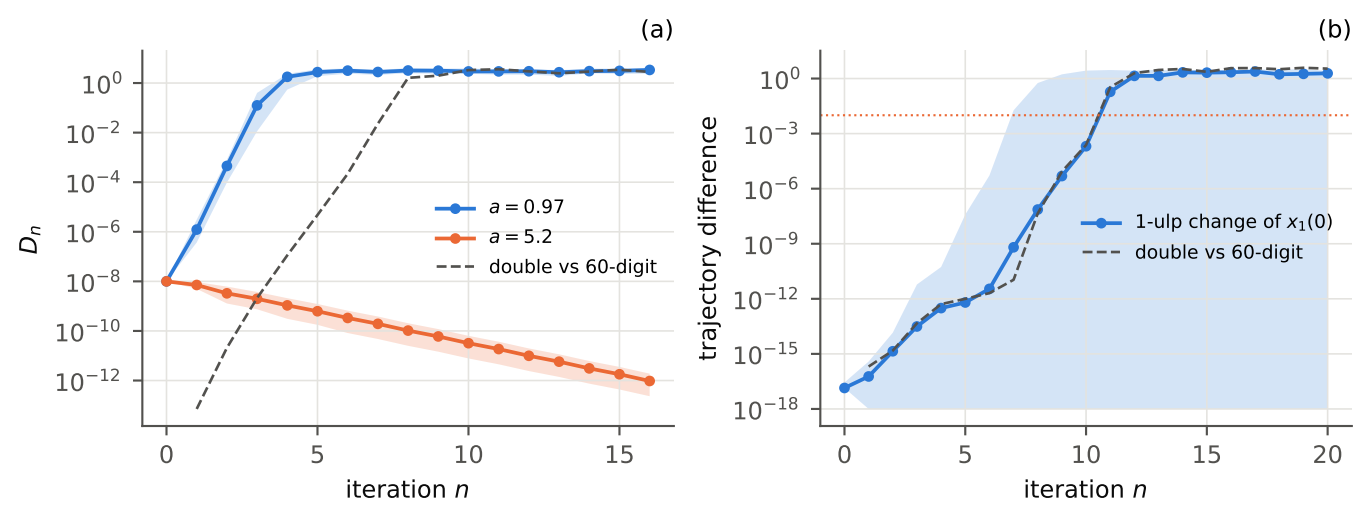
(a)
D_n
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
$a = 0.97$
$a = 5.2$
double vs 60-digit
0
5
10
15
iteration n
(b)
trajectory difference
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
1-ulp change of $x_1(0)$
double vs 60-digit
0
5
10
15
20
iteration n